CASE STUDY: Smart Traffic Signal Control – Value Iteration

CODE:

```
states = ["Low", "Medium", "High"]
actions = ["Short", "Medium", "Long"]
gamma = 0.9
reward = {"Low": 10, "Medium": 5, "High": -10}
V = {s:0 for s in states}
for _ in range(20):
 for s in states:
  V[s] = max(reward[s] + gamma * V[s] for a in actions)
print("Optimal State Values for Traffic Control:", V)
```

OUTPUT:

Optimal State Values for Traffic Control: {'Low': 87.84233454094309, 'Medium': 43.92116727047154, 'High': -87.84233454094309}